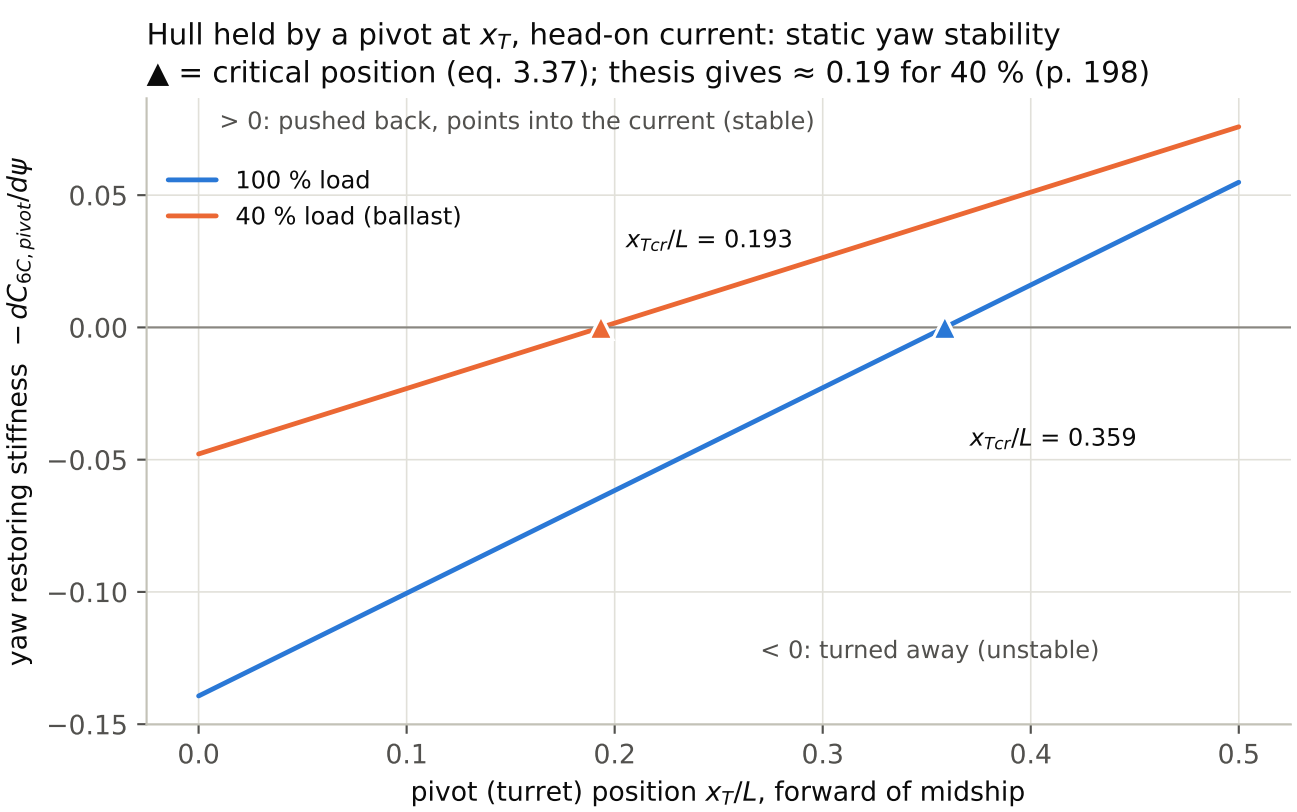
Hull held by a pivot at x_T, head-on current: static yaw stability
▲ = critical position (eq. 3.37); thesis gives ≈ 0.19 for 40 % (p. 198)
> 0: pushed back, points into the current (stable)
100 % load
40 % load (ballast)
x_{Tcr}/L = 0.193
x_{Tcr}/L = 0.359
< 0: turned away (unstable)
yaw restoring stiffness $-dC_{6C,\,pivot}/d\psi$
0.05
0.00
−0.05
−0.10
−0.15
0.0
0.1
0.2
0.3
0.4
0.5
pivot (turret) position x_T/L, forward of midship